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A Prompt-tuning Method For Low-Resource Relation Extraction Based On Enhanced-Prototype

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***Abstract*—Prompt-tuning has been successful in low resource relation extraction. Since there are many relation labels in relation extraction task, it is a challenge to build label verbalizer, which mapping answer word to label space. To address this problem, we propose a prototype construction strategy based on relation semantic enhancement and Prototypical Network, which use attention mechanisms to calculate weighted averages of answer word embedding in support set. We also propose an entity type-enhanced answer word optimization approach to refine the answer word embedding based entity type information in relation context. We conduct experiments on the TACRED and TACRED-Revisit datasets. The experimental results demonstrate that our methods in enhance the accuracy of prototype construction and answer word embeddings.**

Keywords-component; relation extraction; prompt-tuning; low resource;

I. Introduction

Relation Extraction is an essential technique for acquiring knowledge from unstructured text. Deep learning-based methods for relation extraction have shown promising results [1]. However, such methods typically require a substantial amount of annotated data. After the emergence of large-scale pre-trained language models such as BERT and GPT, the methods of pre-training on unlabeled text and fine-tuning on downstream tasks have reached new levels of effectiveness in relation extraction tasks. However, the fine-tuning performance of these models still remains constrained by the quantity of annotated data available. Therefore, many research in relation extraction have shifted towards low-resource scenarios [2]. Prompt-tuning, a novel approach for relation extraction based on pre-training model, utilizes prompt templates and label-word mapping techniques to bridge the gap between pre-trained language models and downstream tasks. Due to the absence of an additional classification layer for pre-trained language models, it is particularly well-suited for low-resource tasks [3].

I love this movie. It was a [MASK] movie.	good → Postivie bad → Negative
A [MASK] news: Tokyo Olympic Daily Preview, July 26th.	technology → Tech sports → Sports science → Science
<e2> Steven Jobs </e2>, co-founder of <e1> Apple <e1>. Steven Jobs [MASK] Apple.	create → per:foundedby found → per:foundedby start → per:foundedby

Figure 1. The challenges associated with Verbalizer in relation extraction.

The Verbalizer aims to establish a connection between the response words generated by the model for prompt input and the task labels. As shown in Figure 1, one of the challenges in relation extraction based on prompt-tuning is the design of suitable label-word mappings. Unlike standard text classification tasks, relation extraction tasks involve a larger number of label classes, each with more complex semantics, making it difficult to build mappings using individual words.

In the past, methods that involved manually or automatically selecting vocabulary from the discrete space of natural language to construct label-word mappings were labor-intensive [4, 5]. Meanwhile, approaches that create virtual label words in the continuous semantic space of pre-trained language models typically require substantial data for training and optimization to achieve optimal results [6, 7], making them challenging to apply directly in low-resource scenarios.

Recent research has incorporated prototype networks into prompt fine-tuning. These methods construct category prototypes by averaging the representations of response words and utilize the Euclidean distance between samples and prototypes to generate classification probabilities. They have shown promising results in low-resource relation extraction tasks. However, existing prompt fine-tuning methods based on prototype networks suffer from the following issues:

1. Simple Prototype Construction: These methods often construct sample prototypes by straightforwardly averaging the response words of the samples. They do not take into account the varying importance of samples in prototype construction, resulting in less robust prototypes that are susceptible to the influence of outlier samples.

2. Neglect of Key Information in Relation Extraction: Relation extraction methods require the consideration of entity and entity type information to enhance classification performance. Existing methods focus on assessing the similarity between representations of response words but overlook the utilization of entity type information.

To address these issues, we propose A Prompt-tuning Method For Low-Resource Relation Extraction Based On Enhanced-Prototype. We introduce semantic information of relation label words into the prototype construction process using attention mechanisms. Additionally, we employ entity type information to perform contrastive learning between support-set and query-set samples to optimize response word embeddings and improve training efficiency.

In summary, this paper makes two key contributions:

1. **Semantics-Based Prototype Construction Strategy**: We introduce a strategy for constructing prototypes based on the semantic information of relation labels. It leverages the Smooth Inverse Frequency (SIF) method to obtain embeddings of relation label words. Furthermore, an attention mechanism is employed between label word embeddings and answer word embeddings to construct prototypes, enhancing the accuracy of prototype representations.

2. **Entity-Type-Based Answer Word Embeddings Enhancement Strategy**: We propose a strategy for optimizing answer word embeddings based on entity type information. It utilizes entity type information to enhance the embeddings of answer words generated by pre-trained language models, thereby enabling the model to generate more accurate answer word embeddings.

We conducted experiments on two publicly available datasets, TACRED and TACRED-Revisit, and the results demonstrate the effectiveness of our approach in low-resource relation extraction.

II. Proposed Method

As shown in Figure 2, the overall approach of this paper can be summarized as follows: First, we construct prompt templates by injecting entity type information into them, creating a prompt input dataset. Next, we use Smooth Inverse Frequency (SIF) [8] method to obtain semantic embeddings for relation labels and use these embeddings to build an attention mechanism by measuring the similarity between label semantic embeddings and answer word embeddings. Finally, we utilize entity types as critical information to perform contrastive learning, further optimizing the answer word embeddings.

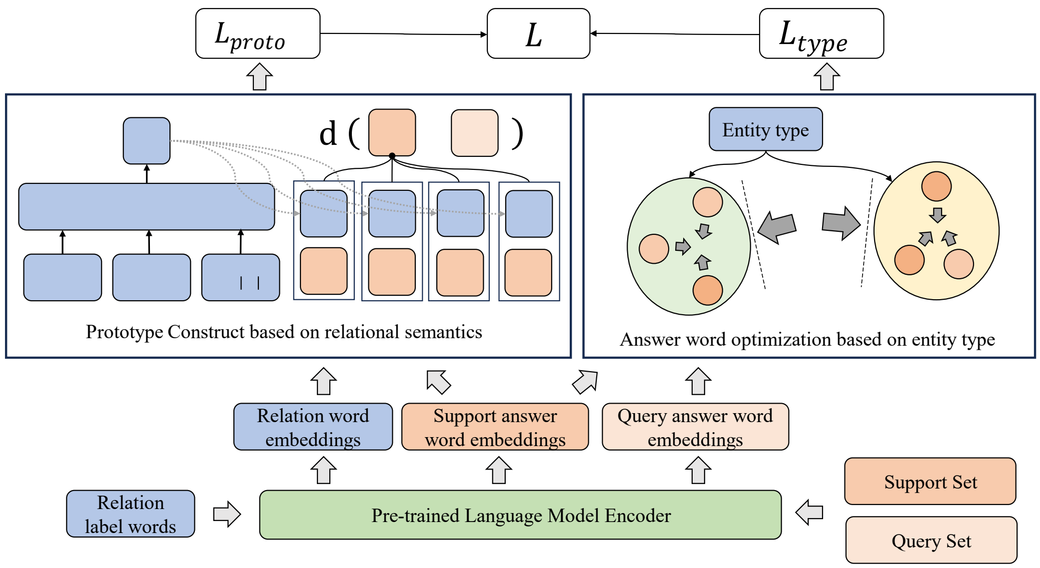

Figure 2. The Framework of our methods.

A. *Prototype Construct based on relational semantic attention mechanism*

The accuracy of prototype network classification relies on prototype representation. Existing prototype-based hint fine-tuning relationship extraction methods adopt a strategy of simple summing and averaging of the support set sample representations when building prototypes. This method is too simple and ignores the differences in importance between samples in the support set for building prototypes, causing the prototype to be very sensitive to abnormal samples. Some methods use attention mechanisms to improve the prototype building process and achieve good results. Inspired by these works, we propose a prototype construct method based on relation semantic attention mechanism.

Diverging from conventional text classification tasks, relation extraction involves labels with rich semantics. Consequently, the semantic content within the relations labels in relation extraction tasks can be regarded as a form of high-quality prior knowledge.

Therefore, we introduce an attention mechanism for prototype construction by measuring the semantic similarity between relationship label words and response words. The objective is to enhance the accuracy of model-generated prototypes.

For a set of relation label words, we split each relation label word into individual words with independent semantics. We then obtain the initial representation of the semantics of each relationship label word by summing the word frequencies weighted by their frequencies:

$$e'_r = \frac{1}{|r|}\sum_{w_j \in r}\frac{a}{a+p(w_j)}e(w_j) \tag{1}$$

We use these initial embeddings of relation label words to form a matrix and apply Principal Component Analysis (PCA) to this matrix to extract its first principal singular vector, which is used to calculate the final embeddings of relation semantics:

$$e_r = e'_r - uu^\top e'_r \tag{2}$$

We calculate weights for each sample using an attention mechanism:

$$h_{c_i} = \frac{1}{|S_{c_i}|} \sum_{S_{c_i} \in S} a_i v_i \tag{3}$$

$$\boldsymbol{a} = softmax\big((W_q H_{[MASK]}) \cdot (W_k e_r{}^{\top})\big) \tag{4}$$

We generate classification probabilities by measuring the distances between answer words embeddings of query-set samples and the prototypes build on support-set samples:

$$p\big(y = k \big| h_{[\mathrm{MASK}]}\big) = \frac{\exp\,(d(h_{[\mathrm{MASK}]}, h_{c_k}))}{\sum_{k'} \exp\,(d\big(h_{[\mathrm{MASK}]}, h_{c_{k'}}\big))} \tag{5}$$

We use cross-entropy to calculate the loss:

$$L_{proto} = -\frac{1}{|Q|} \sum_{h_{[MASK]} \in Q} y \log p\big(y = k \big| h_{[\mathrm{MASK}]}\big) \tag{6}$$

B. Optimization of answer word embeddings based on entity type information

The task of relation extraction differs from ordinary text classification tasks in that it requires considering not only contextual text features but also entity-related features, such as entity type information. Intuitively, pre-trained language models may produce different output response words for the same relation category in response to different prompt inputs. This is because the model's response words are not only related to the relation category of the prompt input but also to the entity types associated with the entity pair in the prompt input. Based on this observation, we propose an entity type-enhanced strategy for optimizing response word representations.

For each sample pair in the training set, we have defined a similarity label based on the relationship label of the sample pair and the entity types:

$$y'_{ij} = \begin{cases} 1, & \text{if } y_i = y_j \text{ and } t_s^i = t_s^j \text{ and } t_o^i = t_o^j \\ 0, & else \end{cases} \tag{7}$$

We optimize the answer word embeddings using the Neighborhood Component Analysis (NCA) loss:

$$L_{type} = \frac{-1}{|D|} \sum_{i \in 1,2,\dots,b} \log \frac{\sum_{\substack{j \in 1,2,\dots,b \\ j \neq i \\ y_{ij}=1}} \exp - |h_i - h_j|}{\sum_{\substack{k \in 1,2,\dots,b \\ k \neq i}} \exp - |h_i - h_k|} \tag{8}$$

Our final loss function is as follows

$$L = L_{proto} + \alpha L_{type} \tag{9}$$

By jointly optimizing the prototype loss based on the semantics of relation label words and the NCA loss of answer words based on entity type information, we hope to further improve the generalization ability of the model in low-resource scenarios.

III. Experiments

A. Datasets

We conduct experiments on two publicly available relation extraction datasets, TACRED and TACRED-Revisit. The TACRED dataset consists of 75,049 relationship instances in the training set, 25,763 instances in the validation set, and 18,659 instances in the test set, encompassing a total of 42 different relationship types. TACRED-Revisit is a dataset constructed based on the original TACRED dataset. TACRED-Revisit retains all samples from the TACRED training set and corrects errors in the development and test sets of TACRED.

B. Experiment setting

To validate the effectiveness of our approach in low-resource relation extraction, we selected representative baseline models for two scenarios: semi-supervised learning and few-shot learning. For the semi-supervised learning scenario, we chose the baseline model MRefG[9], which constructs a reference graph using entity information to establish associations between unlabeled and labeled data, and filters high-quality samples from the unlabeled dataset. For the few-shot learning scenario, we selected the baseline model OntoPrompt[10], which incorporates ontologies from external knowledge graphs as structured knowledge into the prompt templates. We use precision, recall, and F1-score as evaluation metrics.

For the low-resource relation extraction method based on semi-supervised learning, we utilized the TACRED dataset. We performed stratified sampling from the training set to obtain 3%, 10%, and 15% of the training data as labeled data. We also adhered to Li's settings by excluding predictions for the "no_relation" relationship label.

For the low-resource relation extraction method based on few-shot learning, we used the TACRED-Revisit dataset. We conducted few-shot learning experiments with 8, 16, and 32 samples. This involved extracting k instances from each class in the TACRED-Revisit training and validation sets for training and evaluating on the test set.

C. Performance Compare

Table 1. Algorithm Performance In semi-supervised

Methods	**3%**			**10%**			**30%**		
	Precision	***Recall***	***F-1***	***Precision***	***Recall***	***F-1***	***Precision***	***Recall***	***F-1***
BERT	49.8	34.2	41.1	55.7	52.8	54.2	59.1	53.1	56.6
MRefG	56.3	36.3	43.8	59.3	52.0	55.4	61.0	55.6	58.2
Ours	55.6	37.9	45.1	59.0	53.6	56.2	62.6	54.7	58.4

As shown in Table 1, our method in the semi-supervised scenario outperforms the baseline model based on semi-supervised learning with labeled data at 3%, 10%, and 15%. While the MRefG method utilizes entity information to construct a reference graph, mitigating noisy pseudo-labels in unlabeled data and leveraging implicit knowledge in unlabeled data, it tends to focus on shallow text features and does not deeply capture the textual characteristics of relationships. Our approach incorporates relation semantic information to construct label-word mappings and utilizes entity type information to optimize response word embeddings, effectively mapping into the relation semantic knowledge learned by pre-trained language models from unlabeled text. Compared to MRefG, our method is better equipped to capture deeper semantic features of relation extraction tasks.

Table 2. Algorithm Performance In Few-Shot

Methods	*K=8*	*K=16*	*K=32*
BERT	13.5	22.3	28.2
OntoPrompt	28.8	33.1	34.8
Ours	30.1	33.7	35.1

As shown in Table 2, our method surpasses the baseline in few-shot learning scenario with 8, 16, and 32 samples. The baseline method injects ontology knowledge into the model training process by introducing virtual prompt words into the prompt templates. However, this method requires optimizing these virtual prompt words during training, and is most effective in full-supervision scenarios with abundant training data. Nevertheless, in low-resource scenarios, optimizing virtual prompts can lead to overfitting issues. Our approach constructs label word mappings using prototypes, directly optimizing answer word embeddings with task information, without introducing additional learning parameters. Therefore, our method achieves better results in few-shot scenarios.

Table 3. Ablation Experiment

Methods	*K=8*	*K=16*	*K=32*
Ours	30.1	33.7	35.1
Ours w/o SIF	29.0	32.9	34.2
Ours w/o relation	28.7	32.4	33.9
Ours w/o entity	29.7	33.4	34.7

As shown in Table 3, we conducted ablation experiments, which showed that the relational semantic modeling method based on the SIF method, the prototype construction method based on relational semantics, and answer word contrastive learning method based on entity type information all have an impact on our method. Ours w/o SIF method uses the direct addition of label word semantics to replace the process of using the SIP method to obtain relation semantics. This direct addition couples the semantics of tag words together. In the subsequent training process, due to insufficient training data, the semantics of each label word are difficult to decouple during the prototype construction process. In contrast, the SIF method can remove redundant word meanings and accurately model relational semantics, thus improving the effect of prototype construction. The prototype construction method based on relational semantics can use the semantics of relation label words to improve the prototype construction process, reduce the impact of noise information in the support set label word embeddings on the accuracy of prototype construction, and highlight the role of important samples. Finally, the method of using entity type information combined with NCA loss to optimize answer words injects the constraints of entity type information on relation labels into the task of pre-training language models, improving the performance of the model.

IV. Conclusion

We propose a Prompt-tuning Method For Low-Resource Relation Extraction Based On Enhanced-Prototype. This method is based on prototype networks and establishes a relation-based attention mechanism by measuring the semantic similarity between relation label words and answer word embeddings of support-set samples. Additionally, it optimizes the representation of response words by incorporating entity type information into the contrastive learning process between support-set and query-set samples. Experimental results demonstrate that our approach, which leverages designed prompt learning mechanisms to utilize pre-trained language models, outperforms standard semi-supervised learning methods in low-resource relation extraction tasks. Moreover, our method, which constructs label-word mappings using prototypes and optimizes anwser word embeddings with entity type information, is more effective in low-resource scenarios compared to the approach of injecting knowledge into virtual prompts.

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